

AI Recruitment Assistant

Complete 7-Subsystem Architecture & Node Specification Reference

Architecture Summary: The AI Recruitment Assistant automation engine (built on **n8n**, **OpenAI GPT-4o-mini**, **PostgreSQL / Supabase with pgvector**, and **Streamlit**) comprises **81 total nodes** categorized into **7 specialized subsystems**. It manages end-to-end recruitment lifecycle operations: multi-source candidate intake, resume parsing, job requirement extraction, automated fit scoring, multi-channel communications (Gmail/WhatsApp/Google Calendar), semantic vector search (RAG), real-time recruiter analytics, and automated candidate outreach.

Module 1: Candidate Intake & Resume Parsing (11 nodes)

Overview: Captures candidate applications via web forms and file uploads, extracts plain text from resumes, executes LLM-based structured information extraction (skills, experience, education, salary, LinkedIn, GitHub), uploads files to Google Drive, maps relational schema fields, upserts candidate profiles into PostgreSQL, and triggers instant confirmation emails and WhatsApp notifications.

Node Name	Node Type / Version	Functional Role & Execution Logic
Candidate Form	<i>formTrigger (v2.5)</i>	Collects candidate details (Full Name, Email, Phone, PDF resume upload).
PDF Text Extractor	<i>extractFromFile (v1.1)</i>	Extracts raw textual content from the uploaded resume binary buffer.
AI Resume Parser	<i>informationExtractor (LangChain)</i>	Extracts structured attributes: skills, years of experience, companies, education, summary.
OpenAI Model	<i>lmChatOpenAi (GPT-4o-mini)</i>	Powers the resume information extraction model with low-latency structured extraction.
Parse Resume JSON	<i>code</i>	Validates and parses JSON extraction output into standardized property keys.
Upload Resume	<i>googleDrive (v3)</i>	Archives the candidate resume PDF into designated Google Drive storage.
Merge Candidate Data	<i>merge (v3.2)</i>	Combines Google Drive webViewLink with parsed resume attributes and form metadata.
Candidate Data Mapper	<i>set (v3.4)</i>	Normalizes candidate attributes (name, email, skills, currentCompany, location, CTC) for database insertion.
Save Candidate (PostgreSQL)	<i>postgres (upsert)</i>	Upserts candidate record into public.candidates database table.
HR Notification	<i>gmail</i>	Dispatches an application summary email to the recruitment team.
Candidate WhatsApp Confirmation	<i>whatsapp</i>	Sends an immediate automated WhatsApp confirmation message to the applicant.

Module 2: Job Management & Requirement Extraction (10 nodes)

Overview: Enables recruiters to post job requisitions. The AI parser processes raw job descriptions into structured competencies, identifies mandatory vs preferred technical skills, creates target candidate profiles, and generates standard interview questions, storing clean job requisitions in PostgreSQL and notifying the hiring team.

Node Name	Node Type / Version	Functional Role & Execution Logic
HR creates job Form	<i>formTrigger</i>	Recruiter-facing submission form for new job requisitions.
Job Data Mapper	<i>set</i>	Maps incoming form fields (job title, department, raw description, hiring manager).
AI Job Parser	<i>informationExtractor</i>	Analyzes job description to extract structured skills, experience level, and role scope.
OpenAI Chat Model	<i>lmChatOpenAi</i>	LLM reasoning model powering the job requirement breakdown.
Structured Output Parser	<i>outputParserStructured</i>	Enforces strict JSON schema validation for extracted job requirements.
OpenAI Chat Model2	<i>lmChatOpenAi</i>	Secondary model instance for interview question generation.
Merge1	<i>merge</i>	Joins raw form inputs with AI-generated requirement attributes.
Merge Final Job Data	<i>set</i>	Structures final job document for relational storage.
Save Job	<i>postgres (insert)</i>	Inserts structured job vacancy into public.jobs table.
HR Notification1	<i>gmail</i>	Sends a publication confirmation email to the hiring manager.

Module 3: AI Candidate Scoring & Fit Analysis (11 nodes)

Overview: Executes objective algorithmic and generative candidate evaluations by matching candidate qualifications against job requirements. Generates numerical fit scores (0–100), ATS alignment percentages, qualitative recommendation rationales, and strength/gap analysis, persisting results in the applications table.

Node Name	Node Type / Version	Functional Role & Execution Logic
Run Candidate Scoring	<i>manualTrigger</i>	Manual or programmatic execution trigger for candidate scoring runs.
Get Candidate (PostgreSQL)	<i>postgres (select)</i>	Retrieves candidate profile, extracted skills, and background from database.
Get Job (PostgreSQL)	<i>postgres (select)</i>	Retrieves target job specification and required qualifications.
Prepare Candidate Data	<i>set</i>	Formats candidate credentials into evaluation payload.
Prepare Job Data	<i>set</i>	Formats job criteria into evaluation payload.
Merge	<i>merge</i>	Pairs candidate profile with job requirements into unified evaluation context.
AI Candidate Scoring	<i>chainLlm</i>	Evaluates candidate fit against job requirements and computes scores & strengths.
OpenAI Chat Model1	<i>lmChatOpenAi</i>	GPT-4o-mini reasoning engine calculating the match score.
Parse Scoring JSON	<i>code</i>	Parses and validates candidate_score, ats_score, strengths, and recommendations.
Save Application Score	<i>postgres (update)</i>	Updates public.applications record with scores and recommendation summaries.
HR Score Email	<i>gmail</i>	Dispatches candidate assessment scorecard to the recruiter inbox.

Module 4: Application Status & Multi-Channel Communications (17 nodes)

Overview: Orchestrates status lifecycle transitions (Shortlisted, Interview Scheduled, Rejected, Selected). Routes messages through a logic switch, generates dynamic Google Calendar interview invites with meeting links, and sends multi-channel notifications across Gmail and WhatsApp.

Node Name	Node Type / Version	Functional Role & Execution Logic
Webhook - Update Application Status	<i>webhook</i>	Receives status update triggers from Dashboard or external ATS.
Get Application	<i>postgres (select)</i>	Fetches current application state and job linkage.
Get Candidate Contact	<i>postgres (select)</i>	Retrieves candidate email and verified phone number.
Prepare Status Update	<i>set</i>	Constructs status change context and message parameters.
Update Application Status	<i>postgres (update)</i>	Updates stage column in public.applications.
Switch	<i>switch</i>	Routes execution flow based on new status: Shortlist, Interview, Reject, Select.
Shortlist Email	<i>gmail</i>	Dispatches interview preparation and shortlisting notification email.
Send message (WhatsApp)	<i>whatsApp</i>	Sends WhatsApp notification for shortlisted candidates.
Create an event	<i>googleCalendar</i>	Schedules interview on Google Calendar with meeting link and attendees.
Interview Scheduled	<i>gmail</i>	Sends official interview invitation email containing calendar and meeting links.
Send message1 (WhatsApp)	<i>whatsApp</i>	Sends interview date and time reminder via WhatsApp.
Rejection Email Send a message	<i>gmail</i>	Sends respectful and constructive rejection email to candidate.
Send message2 (WhatsApp)	<i>whatsApp</i>	Sends courteous rejection status update via WhatsApp.
Selection Send a message	<i>gmail</i>	Sends job offer / selection letter announcement email.
Send message3 (WhatsApp)	<i>whatsApp</i>	Sends offer announcement alert via WhatsApp.
Internal HR Notification	<i>gmail</i>	Alerts hiring manager of status milestone completion.
Send message4 (WhatsApp)	<i>whatsApp</i>	Sends internal HR status alert via WhatsApp.

Module 5: PGVector Resume Semantic Indexing & Search (RAG) (11 nodes)

Overview: Provides Retrieval-Augmented Generation (RAG) talent discovery. Vectorizes candidate resumes using OpenAI text embeddings, stores vectors in PostgreSQL pgvector, and executes high-speed cosine similarity searches based on recruiter natural language queries.

Node Name	Node Type / Version	Functional Role & Execution Logic
Postgres Trigger	<i>postgresTrigger</i>	Listens for new candidate insertions or updates in PostgreSQL.
Select rows from a table1	<i>postgres (select)</i>	Retrieves candidate resume text for embedding generation.
Edit Fields1	<i>set</i>	Structures candidate document payload with metadata tags (ID, skills, experience).
Embeddings OpenAI (Index)	<i>embeddingsOpenAi</i>	Generates high-dimensional vector embeddings (1536-dim) for resume text.
Resume Document Loader	<i>documentDefaultDataLoader</i>	Loads formatted candidate document chunks into vector store pipeline.
Index Candidate Resume	<i>vectorStorePGVector</i>	Inserts document embeddings into PostgreSQL pgvector index table.
Webhook - Resume Search	<i>webhook</i>	REST endpoint receiving natural language recruiter candidate search queries.
Embeddings OpenAI (Search)	<i>embeddingsOpenAi</i>	Converts recruiter search query into dense query vector.
Search Candidate Resumes	<i>vectorStorePGVector</i>	Executes nearest-neighbor cosine similarity search in pgvector.
Format Resume Search Results	<i>code</i>	Ranks, formats, and deduplicates matching candidates with relevance scores.
Respond Resume Search	<i>respondToWebhook</i>	Returns structured JSON candidate search results to the caller / dashboard.

Module 6: Recruiter Dashboard Real-Time Querying (4 nodes)

Overview: Provides a secured real-time database query bridge between the Streamlit recruiter frontend and PostgreSQL, returning structured candidate pipelines, interview logs, and performance metrics.

Node Name	Node Type / Version	Functional Role & Execution Logic
Recruiter Dashboard Webhook	<i>webhook</i>	Inbound REST webhook trigger invoked by Streamlit dashboard queries.
Query Candidates	<i>postgres (executeQuery)</i>	Executes optimized SQL queries for candidates, stages, and metrics.
Format Dashboard Results	<i>code</i>	Transforms relational query records into dashboard-ready JSON format.
Respond Recruiter Dashboard	<i>respondToWebhook</i>	Returns formatted candidate metrics and pipeline data over HTTP.

Module 7: AI Recruiter Outreach Email Generator (9 nodes)

Overview: Automates hyper-personalized recruiter cold outreach. Combines candidate background highlights with target job requirements to generate tailored, high-conversion recruitment emails, sends them via Gmail, and logs the communication.

Node Name	Node Type / Version	Functional Role & Execution Logic
Generate Outreach Email Webhook	<i>webhook</i>	Trigger receiving candidate ID, target job ID, and tone specifications.
Get Candidate For Outreach	<i>postgres (select)</i>	Pulls candidate specific achievements, previous companies, and skills.
Get Job For Outreach	<i>postgres (select)</i>	Pulls target role requirements, benefits, and company mission.
Merge Outreach Data	<i>merge</i>	Combines candidate and job records into unified prompt context.
OpenAI Chat Model (Outreach)	<i>lmChatOpenAi</i>	Generates personalized, engaging outreach copy.
Structured Output Parser (Outreach)	<i>outputParserStructured</i>	Enforces structured output schema (subject line, body, CTA).
AI Outreach Email Generator	<i>agent</i>	Assembles final tailored message with personalized hooks.
Send Outreach Email	<i>gmail</i>	Dispatches outreach email directly to candidate email inbox.
Respond Outreach Email	<i>respondToWebhook</i>	Returns generated email subject and body back to dashboard UI.